

**Name : Sanika Jumde**

**Sem : V**

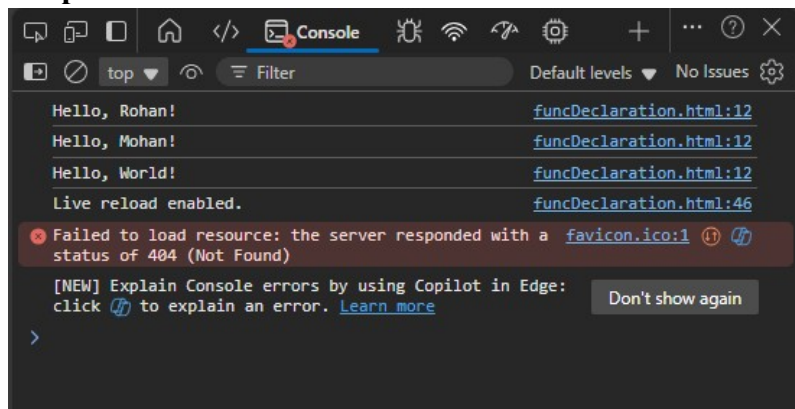
**PRN: 23070521225**

## Easy Level Code

### Code 1:

```
funcDeclaration.html > ...
1 | |!-- Function Declaration:
2 | |Write a function that takes a name and prints "Hello, <name>!".
3 | |Call it with different names. -->
4 | <!DOCTYPE html>
5 | <head>
6 | |<title>Welcome Message</title>
7 | </head>
8 | <body>
9 | |<script>
10 | |    function displayName(name)
11 | |    {
12 | |        console.log("Hello, ${name}!");
13 | |    }
14 | |    displayName("Rohan");
15 | |    displayName("Mohan");
16 | |    displayName("World");
17 | |</script>
18 | </body>
19 | </html>
```

### Output:

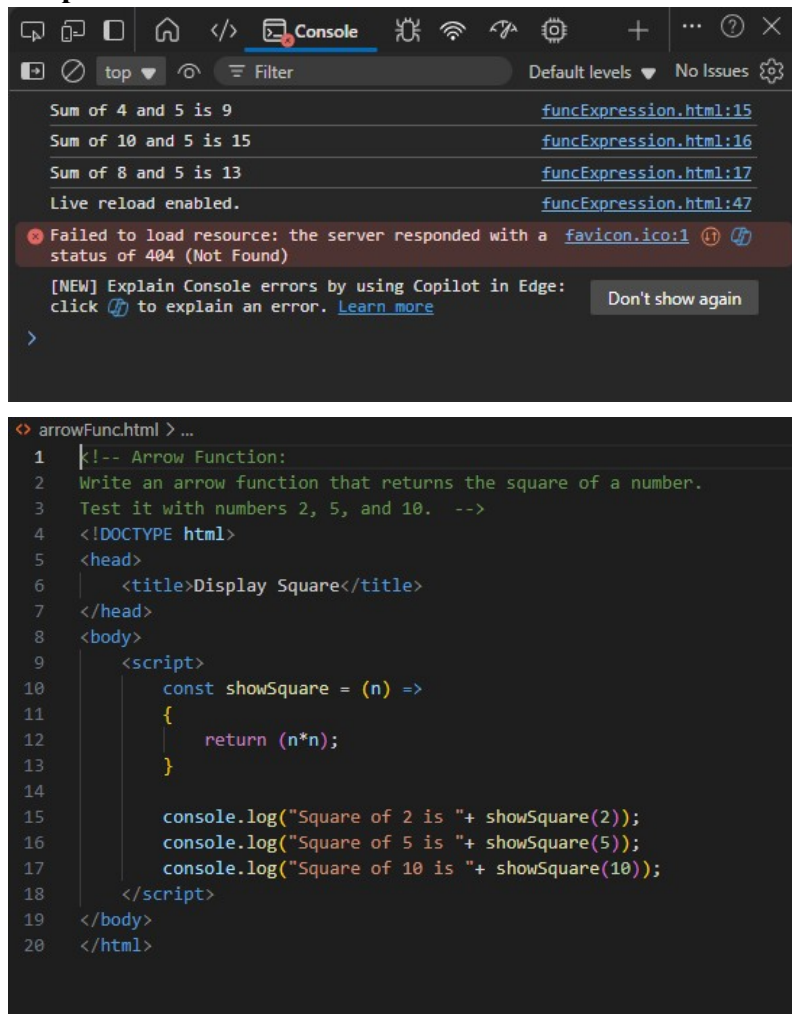


```
top Filter Default levels No Issues
Hello, Rohan! funcDeclaration.html:12
Hello, Mohan! funcDeclaration.html:12
Hello, World! funcDeclaration.html:12
Live reload enabled. funcDeclaration.html:46
Failed to load resource: the server responded with a status of 404 (Not Found)
[NEW] Explain Console errors by using Copilot in Edge:
click to explain an error. Learn more Don't show again
```

```
funcExpression.html > ...
1 | |!-- Function Expression:
2 | |Create a function expression that takes two numbers and returns their sum.
3 | |Test it with at least 3 sets of numbers. -->
4 | <!DOCTYPE html>
5 | <head>
6 | |<title>Addition</title>
7 | </head>
8 | <body>
9 | |<script>
10 | |    const sum = function(a, b)
11 | |    {
12 | |        return (a+b);
13 | |    }
14 | |
15 | |    console.log("Sum of 4 and 5 is " + sum(4,5));
16 | |    console.log("Sum of 10 and 5 is " + sum(10,5));
17 | |    console.log("Sum of 8 and 5 is " + sum(8,5));
18 | |</script>
19 | </body>
20 | </html>
```

## Code 2:

### Output:



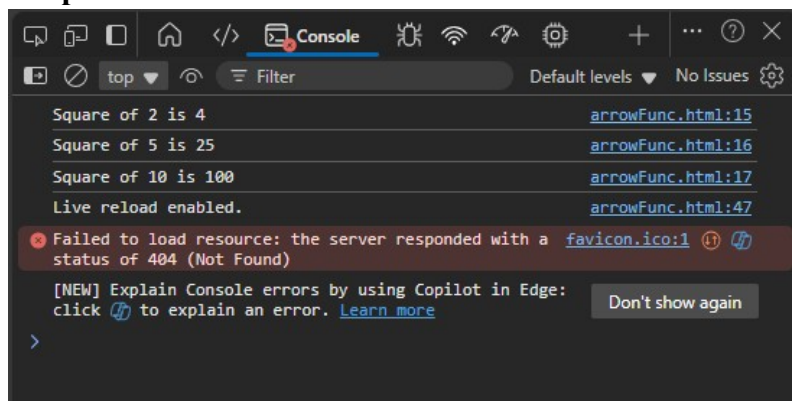
The screenshot shows the VS Code interface. The Console panel at the top displays the following output:

```
Sum of 4 and 5 is 9                                     funcExpression.html:15
Sum of 10 and 5 is 15                                    funcExpression.html:16
Sum of 8 and 5 is 13                                    funcExpression.html:17
Live reload enabled.                                    funcExpression.html:47
Failed to load resource: the server responded with a status of 404 (Not Found) favicon.ico:1
[NEW] Explain Console errors by using Copilot in Edge: click to explain an error. Learn more Don't show again
```

The Editor panel below shows the code for `arrowFunc.html`:

```
1 | <!-- Arrow Function:
2 | Write an arrow function that returns the square of a number.
3 | Test it with numbers 2, 5, and 10. -->
4 | <!DOCTYPE html>
5 | <head>
6 | | <title>Display Square</title>
7 | </head>
8 | <body>
9 | | <script>
10 | |   const showSquare = (n) =>
11 | |   {
12 | |     return (n*n);
13 | |   }
14 | |
15 | |   console.log("Square of 2 is "+ showSquare(2));
16 | |   console.log("Square of 5 is "+ showSquare(5));
17 | |   console.log("Square of 10 is "+ showSquare(10));
18 | | </script>
19 | </body>
20 | </html>
```

### Output:



The screenshot shows the VS Code interface. The Console panel at the top displays the following output:

```
Square of 2 is 4                                         arrowFunc.html:15
Square of 5 is 25                                        arrowFunc.html:16
Square of 10 is 100                                     arrowFunc.html:17
Live reload enabled.                                    arrowFunc.html:47
Failed to load resource: the server responded with a status of 404 (Not Found) favicon.ico:1
[NEW] Explain Console errors by using Copilot in Edge: click to explain an error. Learn more Don't show again
```

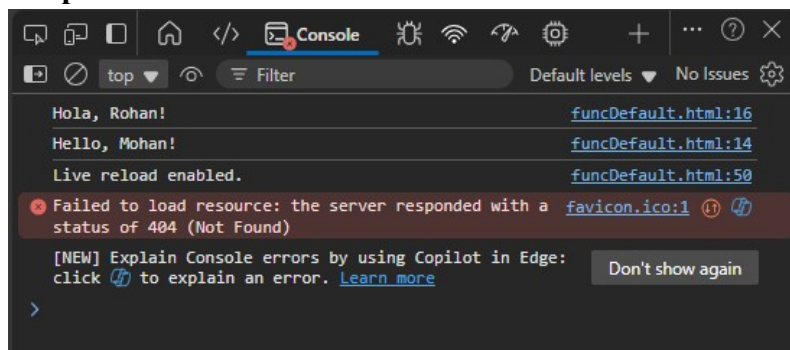
The Editor panel below shows the code for `arrowFunc.html`:

```
1 | <!-- Arrow Function:
2 | Write an arrow function that returns the square of a number.
3 | Test it with numbers 2, 5, and 10. -->
4 | <!DOCTYPE html>
5 | <head>
6 | | <title>Display Square</title>
7 | </head>
8 | <body>
9 | | <script>
10 | |   const showSquare = (n) =>
11 | |   {
12 | |     return (n*n);
13 | |   }
14 | |
15 | |   console.log("Square of 2 is "+ showSquare(2));
16 | |   console.log("Square of 5 is "+ showSquare(5));
17 | |   console.log("Square of 10 is "+ showSquare(10));
18 | | </script>
19 | </body>
20 | </html>
```

### Code 3:

```
<> funcDefault.html > ...
1  <!-- Function Declaration with Default Parameter:
2  Write a function that takes a name and a language (default &quot;English&quot;) and
3  prints &quot;Hello&quot; in that language.
4  Example: (&quot;Alice&quot;, &quot;Spanish&quot;)+ &quot;Hola, Alice!&quot;. -->
5  <!DOCTYPE html>
6  <head>
7  |   <title>Default Parameters</title>
8  </head>
9  <body>
10 |   <script>
11 |     function showLanguage(name, language = "English")
12 |     {
13 |       if(language == "English")
14 |       {
15 |         console.log(`Hello, ${name}!`);
16 |       }
17 |       else if(language == "Spanish")
18 |       {
19 |         console.log(`Hola, ${name}!`);
20 |       }
21 |     }
22 |     showLanguage("Rohan", "Spanish");
23 |     showLanguage("Mohan");
24 |   </script>
25 </body>
26 </html>
```

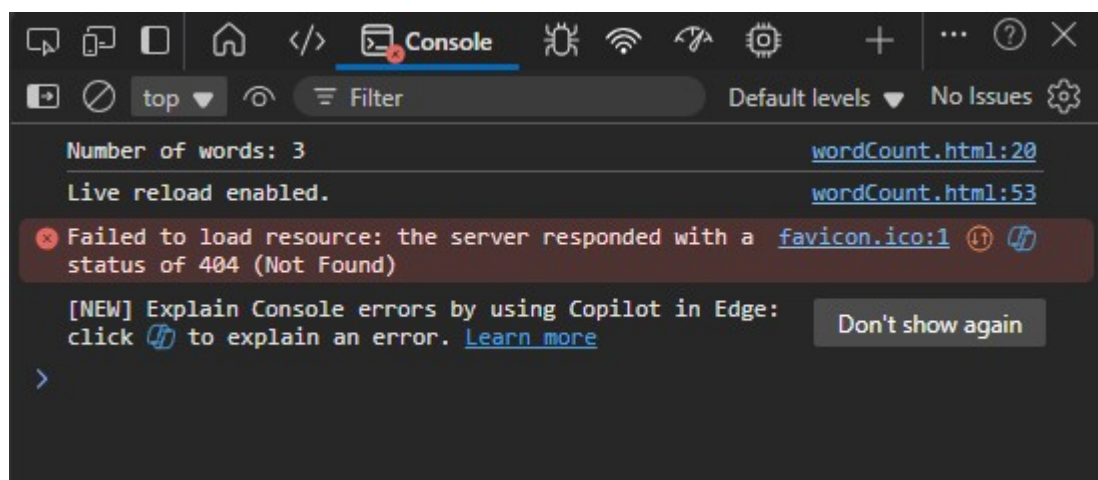
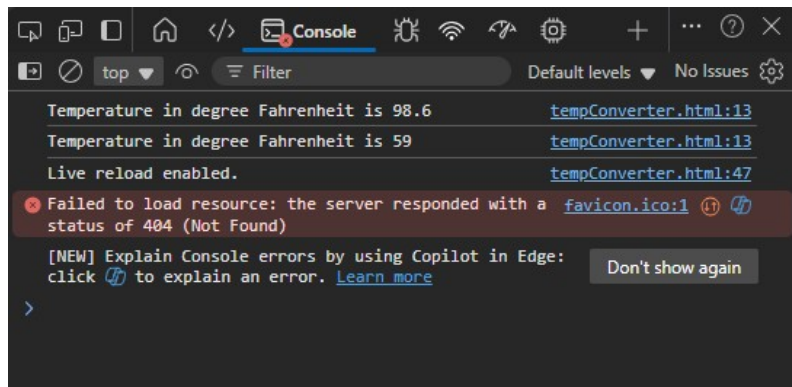
### Output:



```
<> tempConverter.html > ...
1  <!-- Function Expression - Simple Math:
2  Create a function expression that converts Celsius to Fahrenheit.
3  Formula: (C x 9/5) + 32. -->
4  <!DOCTYPE html>
5  <head>
6  |   <title>Temperature Conversion</title>
7  </head>
8  <body>
9  |   <script>
10 |     function celsiusToFahrenheit(tc)
11 |     {
12 |       let tf = (tc * 9 / 5) + 32;
13 |       console.log('Temperature in degree Fahrenheit is '+ tf);
14 |     }
15 |
16 |     celsiusToFahrenheit(37);
17 |     celsiusToFahrenheit(15);
18 |   </script>
19 </body>
20 </html>
```

### Output:

## Code 4:

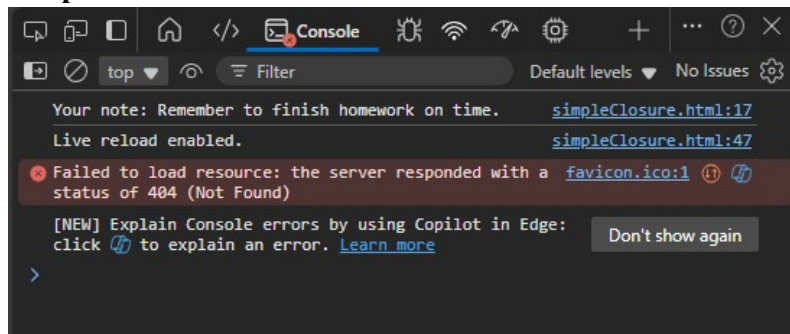


## Output:

## Code 5:

```
simpleClosure.html > ...
1  |<!-- Simple Closure - Personal Note:
2  |  Write a closure that stores a note and returns a function to read it later. -->
3  |  <!DOCTYPE html>
4  |  <head>
5  |    <title>Personal Note</title>
6  |  </head>
7  |  <body>
8  |    <script>
9  |      function createNote(note)
10 |      {
11 |        return function(){
12 |          return `Your note: ${note}`;
13 |        };
14 |      }
15 |
16 |      const myNote = createNote("Remember to finish homework on time.");
17 |      console.log(myNote());
18 |    </script>
19 |  </body>
20 | </html>
```

## Output:

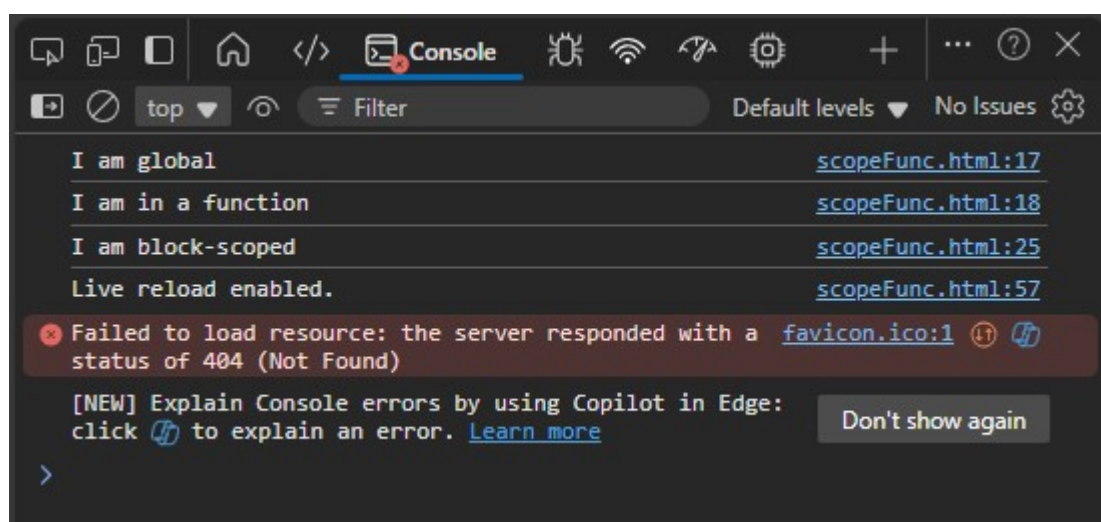


## Moderate Level Code

### Code 1:

```
<?scopeFunc.html> > ...
1  <!-- Scope Practice:
2  Create a global variable and a block-scoped variable.
3  Demonstrate which ones are accessible inside and outside a function and a block
4  { }. -->
5  <!DOCTYPE html>
6  <head>
7  <title>Display Scope</title>
8  </head>
9  <body>
10 <script>
11     //global variable
12     let globalVar = "I am global";
13     function testScope()
14     {
15         //function variable or local variable
16         let functionVar = "I am in a function";
17         console.log(globalVar); // Accessible
18         console.log(functionVar); // Accessible
19     }
20     testScope();
21     // console.log(functionVar); // Error
22     {
23         //block variable
24         let blockVar = "I am block-scoped";
25         console.log(blockVar); // Accessible
26     }
27     // console.log(blockVar); // Error
28 </script>
29 </body>
30 </html>
```

### Output:





## Code 2:

```
stringEnd.html > ...
1  |<!-- Arrow Function String Check:
2  |  Create an arrow function that checks if a sentence ends with &quot;!&quot;.
3  |  Test it with both true and false cases.    -->
4  |<!DOCTYPE html>
5  |<head>
6  |  <title>Check String End</title>
7  |</head>
8  |<body>
9  |  <script>
10 |      const endsWithExclamation = (sentence) =>
11 |      {
12 |          let lastChar = sentence[sentence.length - 1];
13 |          return lastChar === "!";
14 |      }
15 |
16 |      console.log("'Hello!': " + endsWithExclamation("Hello!"));
17 |      console.log("'Hello': " + endsWithExclamation("Hello"));
18 |      console.log("'Wow!!!': " + endsWithExclamation("Wow!!!"));
19 |      console.log("'No way': " + endsWithExclamation("No way"));
20 |  </script>
21 |</body>
22 |</html>
```

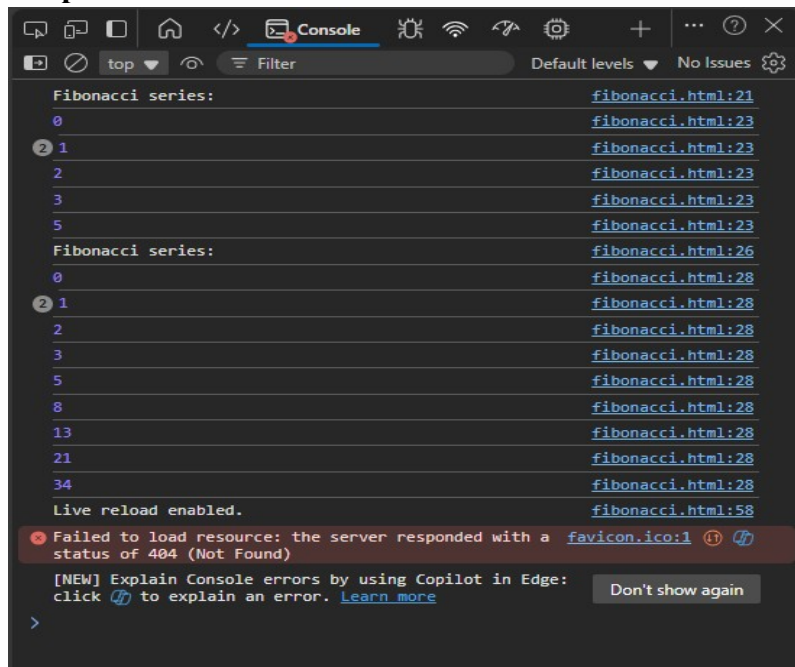
## Output:

```
top Filter Default levels No Issues
'Hello!': true stringEnd.html:13
'Hello': false stringEnd.html:14
'Wow!!!': true stringEnd.html:15
'No way': false stringEnd.html:16
Live reload enabled. stringEnd.html:46
Failed to load resource: the server responded with a status of 404 (Not Found)
[NEW] Explain Console errors by using Copilot in Edge:
click to explain an error. Learn more
>
```

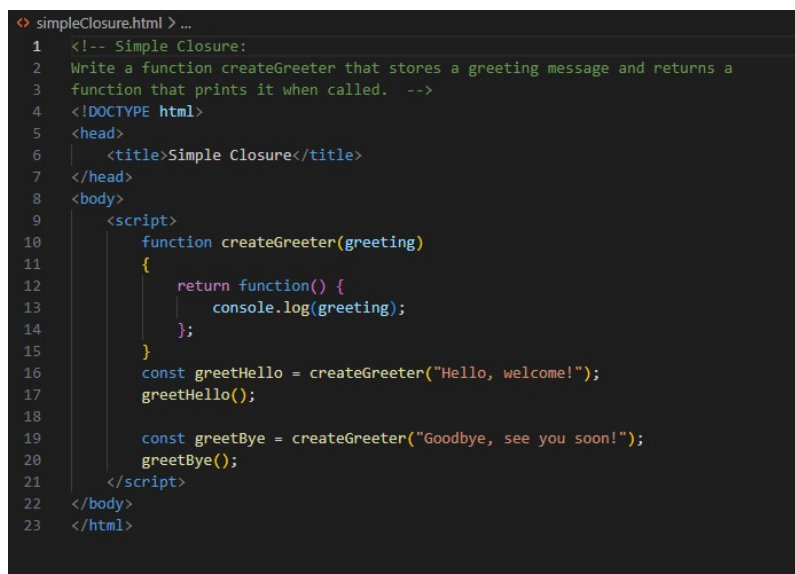
```
fibonacci.html > ...
1  |<!-- Function Expression + Recursion:
2  |  Write a named function expression that finds the nth Fibonacci number.
3  |  Test with n = 6 and n = 10.    -->
4  |<!DOCTYPE html>
5  |<head>
6  |  <title>Fibonacci Series</title>
7  |</head>
8  |<body>
9  |  <script>
10 |      const fibonacci = function(n)
11 |      {
12 |          if(n==0)
13 |              return 0;
14 |          else if(n==1)
15 |              return 1;
16 |          else
17 |              return fibonacci(n-1) + fibonacci(n-2);
18 |      }
19 |
20 |      n = 6;
21 |      console.log("Fibonacci series: ");
22 |      for(let x=0; x<n; x++)
23 |          console.log(fibonacci(x));
24 |
25 |      n = 10;
26 |      console.log("Fibonacci series: ");
27 |      for(let x=0; x<n; x++)
28 |          console.log(fibonacci(x));
29 |  </script>
30 |</body>
31 |</html>
```

### Code 3:

### Output:

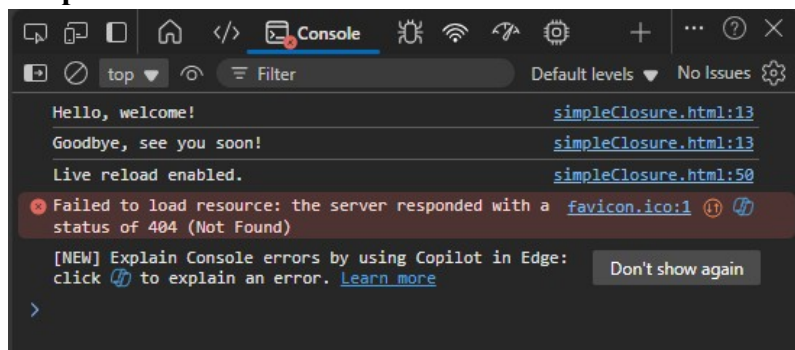


```
Fibonacci series: fibonacci.html:21
0 fibonacci.html:23
1 fibonacci.html:23
2 fibonacci.html:23
3 fibonacci.html:23
5 fibonacci.html:23
Fibonacci series: fibonacci.html:26
0 fibonacci.html:28
1 fibonacci.html:28
2 fibonacci.html:28
3 fibonacci.html:28
5 fibonacci.html:28
8 fibonacci.html:28
13 fibonacci.html:28
21 fibonacci.html:28
34 fibonacci.html:28
Live reload enabled. fibonacci.html:58
Failed to load resource: the server responded with a status of 404 (Not Found)
[NEW] Explain Console errors by using Copilot in Edge:
click to explain an error. Learn more
Don't show again
```



```
simpleClosure.html > ...
1 <!-- Simple Closure:
2 Write a function createGreeter that stores a greeting message and returns a
3 function that prints it when called. -->
4 <!DOCTYPE html>
5 <head>
6   <title>Simple Closure</title>
7 </head>
8 <body>
9   <script>
10     function createGreeter(greeting)
11     {
12       return function() {
13         console.log(greeting);
14       };
15     }
16     const greetHello = createGreeter("Hello, welcome!");
17     greetHello();
18
19     const greetBye = createGreeter("Goodbye, see you soon!");
20     greetBye();
21   </script>
22 </body>
23 </html>
```

### Output:



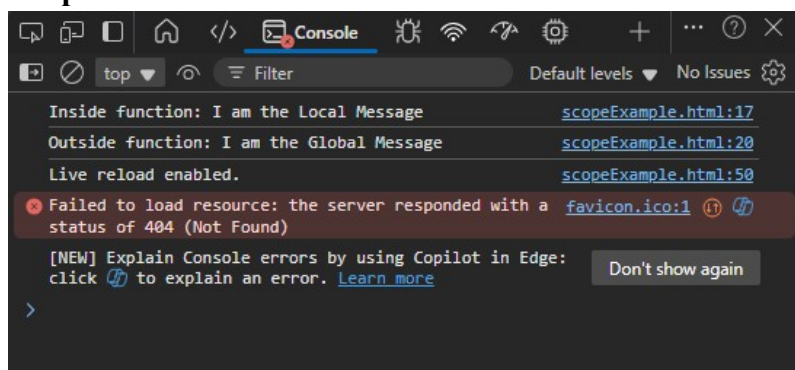
```
Hello, welcome! simpleClosure.html:13
Goodbye, see you soon! simpleClosure.html:13
Live reload enabled. simpleClosure.html:50
Failed to load resource: the server responded with a status of 404 (Not Found)
[NEW] Explain Console errors by using Copilot in Edge:
click to explain an error. Learn more
Don't show again
```



## Code 4:

```
scopeExample.html > ...
1  <!-- Scope Example - Variable Shadowing:
2  Create a function where a local variable has the same name as a global variable.
3  Show how JavaScript chooses which one to use. -->
4  <!DOCTYPE html>
5  <head>
6  |   <title>Variable Shadowing</title>
7  </head>
8  <body>
9  |   <script>
10 |       // Global variable
11 |       let message = "I am the Global Message";
12 |
13 |       function showMessage()
14 |       {
15 |           //Local variable
16 |           let message = "I am the Local Message";
17 |           console.log("Inside function:", message);
18 |       }
19 |       showMessage();
20 |       console.log("Outside function:", message);
21 |   </script>
22 </body>
23 </html>
```

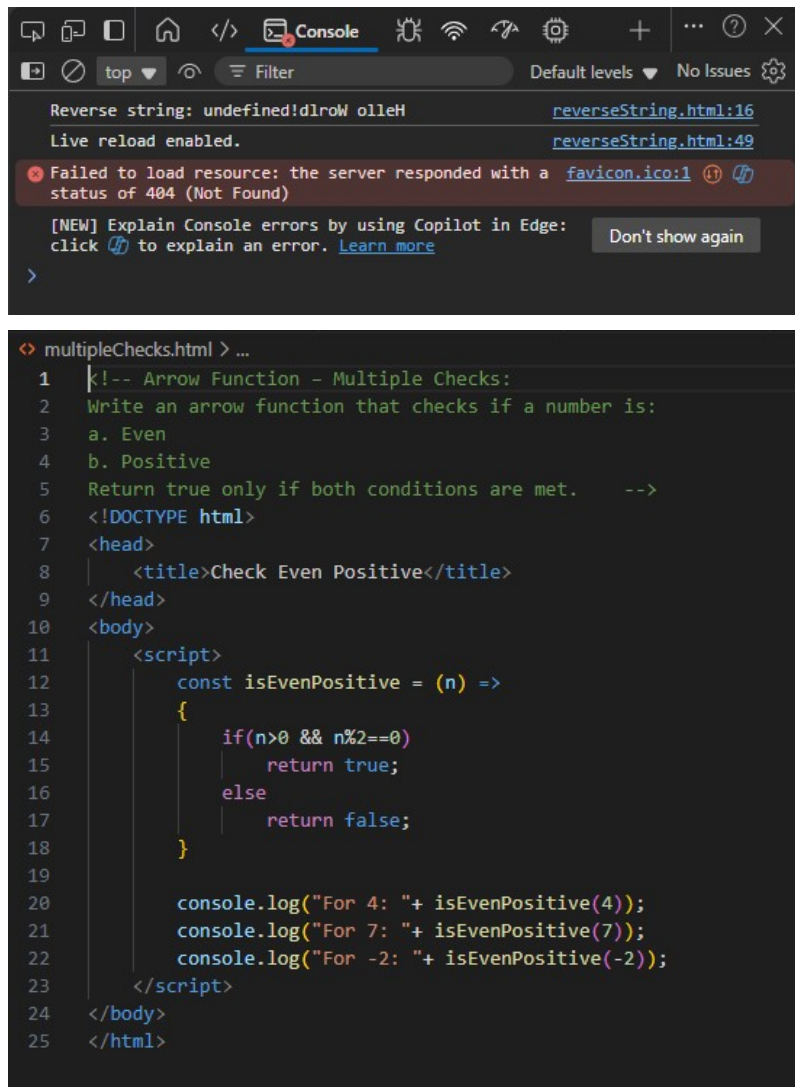
## Output:



```
reverseString.html > ...
1  <!-- Function Expression - Reverse String:
2  Make a function expression that reverses a string without using built-in .reverse(). -->
3  <!DOCTYPE html>
4  <head>
5  |   <title>Display Reverse String</title>
6  </head>
7  <body>
8  |   <script>
9  |       const reverseString = function(str)
10 |       {
11 |           let rev = "";
12 |           for(let x=str.length; x>=0; x--)
13 |           {
14 |               rev += str[x];
15 |           }
16 |           console.log("Reverse string: "+ rev);
17 |       }
18 |
19 |       reverseString("Hello World!");
20 |   </script>
21 </body>
22 </html>
```

## Output:

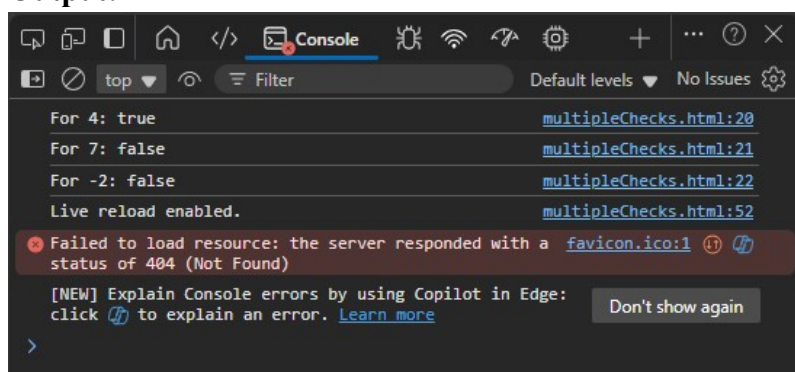
## Code 5:



```
Reverse string: undefined!dlroW olleH reverseString.html:16
Live reload enabled. reverseString.html:49
Failed to load resource: the server responded with a status of 404 (Not Found) favicon.ico:1
[NEW] Explain Console errors by using Copilot in Edge: click to explain an error. Learn more Don't show again
>

multipleChecks.html > ...
1  <!-- Arrow Function - Multiple Checks:
2  Write an arrow function that checks if a number is:
3  a. Even
4  b. Positive
5  Return true only if both conditions are met. -->
6  <!DOCTYPE html>
7  <head>
8  |   <title>Check Even Positive</title>
9  </head>
10 <body>
11 |   <script>
12 |       const isEvenPositive = (n) =>
13 |       {
14 |           if(n>0 && n%2==0)
15 |               return true;
16 |           else
17 |               return false;
18 |       }
19 |
20 |       console.log("For 4: " + isEvenPositive(4));
21 |       console.log("For 7: " + isEvenPositive(7));
22 |       console.log("For -2: " + isEvenPositive(-2));
23 |   </script>
24 </body>
25 </html>
```

## Output:

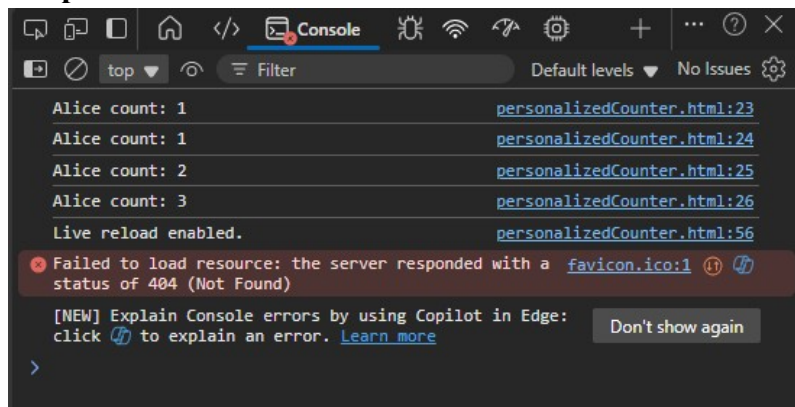


```
For 4: true multipleChecks.html:20
For 7: false multipleChecks.html:21
For -2: false multipleChecks.html:22
Live reload enabled. multipleChecks.html:52
Failed to load resource: the server responded with a status of 404 (Not Found) favicon.ico:1
[NEW] Explain Console errors by using Copilot in Edge: click to explain an error. Learn more Don't show again
>
```

## Code 6:

```
personalizedCounter.html > ...
1  <!-- Closure - Personalized Counter:
2  Create a closure that counts how many times a specific person's name is entered.
3  Example: If "Alice" is entered three times, it should return 3. -->
4  <!DOCTYPE html>
5  <head>
6  |   <title>Personalized Counter</title>
7  </head>
8  <body>
9  |   <script>
10 |       function createNameCounter(personName)
11 |       {
12 |           let count = 0;
13 |
14 |           return function(name) {
15 |               if (name === personName)
16 |                   count++;
17 |               return `${personName} count: ${count}`;
18 |           };
19 |       }
20 |
21 |       const countAlice = createNameCounter("Alice");
22 |
23 |       console.log(countAlice("Alice")); // Alice count: 1
24 |       console.log(countAlice("Bob"));   // Alice count: 1 (no change)
25 |       console.log(countAlice("Alice")); // Alice count: 2
26 |       console.log(countAlice("Alice")); // Alice count: 3
27 |   </script>
28 </body>
29 </html>
```

## Output:



Alice count: 1 [personalizedCounter.html:23](#)

Alice count: 1 [personalizedCounter.html:24](#)

Alice count: 2 [personalizedCounter.html:25](#)

Alice count: 3 [personalizedCounter.html:26](#)

Live reload enabled. [personalizedCounter.html:56](#)

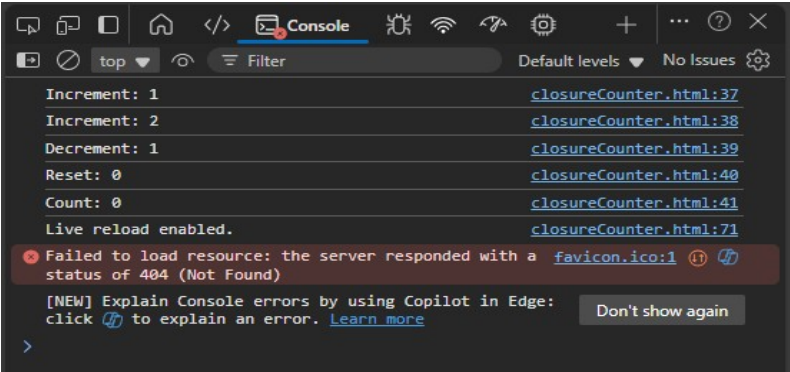
Failed to load resource: the server responded with a status of 404 (Not Found) [favicon.ico:1](#)

[NEW] Explain Console errors by using Copilot in Edge: click [🔗](#) to explain an error. [Learn more](#) [Don't show again](#)

# Advance Level Code

## Code 1:

```
closureCounter.html > ...
1  |<!-- Closure for Counter:
2  |Create a closure-based counter that allows incrementing, decrementing, and
3  |resetting the count.
4  |Each operation should be a separate function. -->
5  |
6  |<!DOCTYPE html>
7  |<head>
8  |  <title>Closure Counter</title>
9  |</head>
10 |<body>
11 |  <script>
12 |    function createCounter()
13 |    {
14 |      let count = 0;
15 |
16 |      return{
17 |        increment: function(){
18 |          count++;
19 |          return count;
20 |        },
21 |        decrement: function(){
22 |          count--;
23 |          return count;
24 |        },
25 |        reset: function(){
26 |          count = 0;
27 |          return count;
28 |        },
29 |        getCount: function(){
30 |          return count;
31 |        }
32 |      };
33 |    }
34 |
35 |    const counter = createCounter();
36 |
37 |    console.log("Increment: "+ counter.increment()); // 1
38 |    console.log("Increment: "+ counter.increment()); // 2
39 |    console.log("Decrement: "+ counter.decrement()); // 1
40 |    console.log("Reset: "+ counter.reset()); // 0
41 |    console.log("Count: "+ counter.getCount()); // 0
42 |  </script>
43 |</body>
44 |</html>
```



## Code 2:

## Output:

```
<> mixedFunc.html > ...
1  <!-- Mixed Function Types:
2  Function Declaration: Generates a random number.
3  Function Expression: Checks if the number is prime.
4  Arrow Function: Prints &quot;Prime&quot; or &quot;Not Prime&quot; based on the result.
5  Use all three in sequence.      -->
6  <!DOCTYPE html>
7  <head>
8  |   <title>Mixed Function Types</title>
9  </head>
10 <body>
11 |   <script>
12 |     function generateRandomNumber(limit = 50)
13 |     {
14 |       return Math.floor(Math.random() * limit) + 1;
15 |     }
16 |
17 |     const isPrime = function(n)
18 |     {
19 |       if(n<2)
20 |         return false;
21 |       for(let x=2; x<=n/2; x++)
22 |         if(n%x === 0)
23 |           return false;
24 |       return true;
25 |     };
26 |
27 |     const printPrimeStatus = (num, primeCheck) =>
28 |     {
29 |       console.log(`${num} → ${primeCheck ? "Prime" : "Not Prime"}`);
30 |     };
31 |
32 |     let randomNum = generateRandomNumber();
33 |     let primeCheck = isPrime(randomNum);
34 |     printPrimeStatus(randomNum, primeCheck);
35 |   </script>
36 </body>
37 </html>
```



## Output:

```
1 1 bankClosure.html > ...
2 |{-- Private Data with Closure:
3 |Make a createBankAccount function that stores a balance.
4 |Provide deposit, withdraw, and checkBalance methods.
5 |Ensure balance can't be accessed directly from outside. -->
6 <!DOCTYPE html>
7 <head>
8   <title>Bank Closure</title>
9 </head>
10 <body>
11   <script>
12     function createBankAccount(initialBalance = 0)
13     {
14       let balance = initialBalance; // private variable, not accessible outside the function block
15
16       return{
17         deposit: function(amount){
18           if(amount>0)
19           {
20             balance += amount;
21             return 'Deposited: ${amount}, New Balance: ${balance}';
22           }
23           return "Deposit amount must be positive!";
24         },
25         withdraw: function(amount){
26           if(amount>balance)
27             return "Insufficient funds!";
28           if(amount>0)
29           {
30             balance -= amount;
31             return 'Withdrew: ${amount}, New Balance: ${balance}';
32           }
33           return "Withdraw amount must be positive!";
34         },
35         checkBalance: function(){
36           return 'Current Balance: ${balance}';
37         }
38       };
39
40       const myAccount = createBankAccount(100);
41
42       console.log(myAccount.checkBalance()); // Current Balance: 100
43       console.log(myAccount.deposit(50)); // Deposited: 50, New Balance: 150
44       console.log(myAccount.withdraw(70)); // Withdrew: 70, New Balance: 80
45       console.log(myAccount.withdraw(200)); // Insufficient funds!
46       console.log(myAccount.checkBalance()); // Current Balance: 80
47
48       console.log(myAccount.balance); // undefined
49     }
50   </script>
51 </body>
52 </html>
```

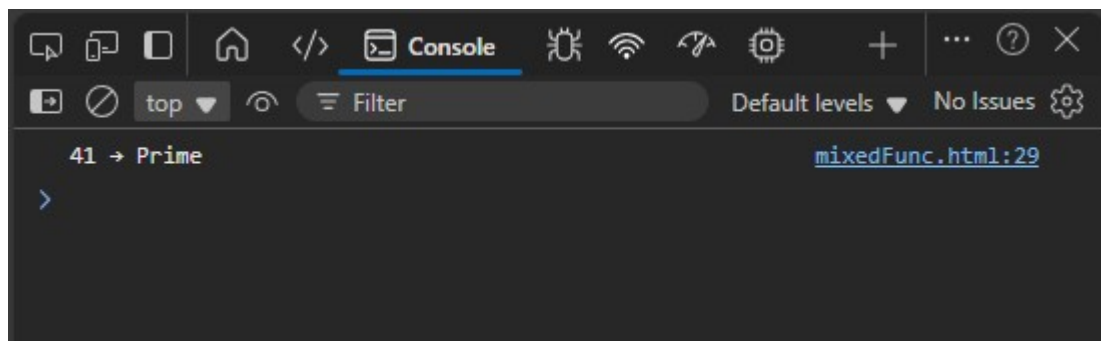
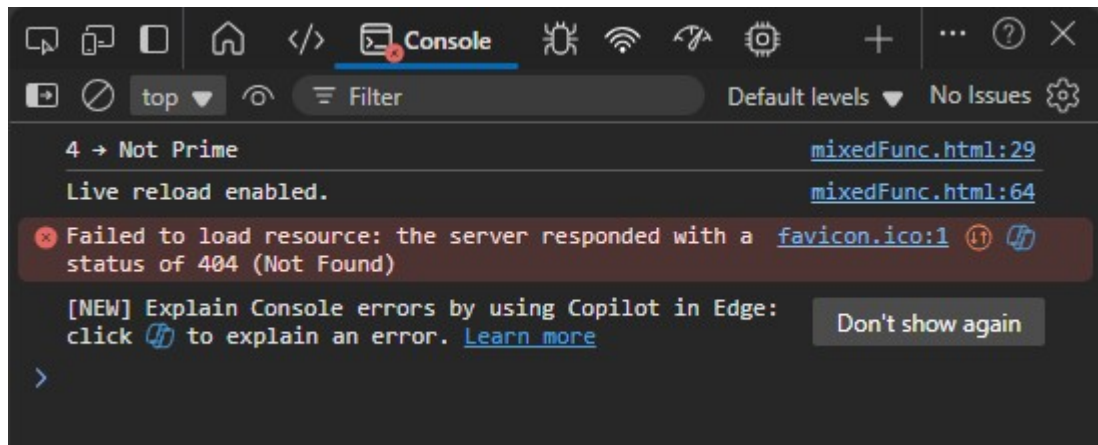
Current Balance: 100 [bankClosure.html:42](#)  
Deposited: 50, New Balance: 150 [bankClosure.html:43](#)  
Withdrew: 70, New Balance: 80 [bankClosure.html:44](#)  
Insufficient funds! [bankClosure.html:45](#)  
Current Balance: 80 [bankClosure.html:46](#)  
undefined [bankClosure.html:48](#)  
Live reload enabled. [bankClosure.html:78](#)

Failed to load resource: the server responded with a status of 404 (Not Found) [favicon.ico:1](#)

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## Code 3:

## Output:



## Output:

### Code 4:

```
<> scopeChain.html > ...
1  <!-- Scope Chain Debugging:
2  Write code with three nested functions (outer, middle, inner) each having variables
3  with the same name.
4  Log which one gets accessed in the innermost function and explain why.    -->
5  <!DOCTYPE html>
6  <head>
7  |   <title>Scope Chain Debugging</title>
8  </head>
9  <body>
10 |   <script>
11 |       function outer()
12 |       {
13 |           let value = "Outer Scope";
14 |
15 |           function middle()
16 |           {
17 |               let value = "Middle Scope";
18 |
19 |               function inner()
20 |               {
21 |                   let value = "Inner Scope";
22 |                   console.log("Accessed in inner:", value);
23 |               }
24 |               inner();
25 |           }
26 |           middle();
27 |       }
28 |       outer();
29 |   </script>
30 </body>
31 </html>
```

## Output:

